04-03-2026

RANGE

In [1]:
```
range()
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
Cell In[1], line 1
----> 1 range()

TypeError: range expected at least 1 argument, got 0
```

In [2]:
```
range(5)
```

Out[2]:
```
range(0, 5)
```

In [3]:
```
list(range(5))
```

Out[3]:
```
[0, 1, 2, 3, 4]
```

In [4]:
```
list(range(1,5))
```

Out[4]:
```
[1, 2, 3, 4]
```

In [5]:
```
list(range(1,10,3))
```

Out[5]:
```
[1, 4, 7]
```

In [6]:
```
list(range(1,10,5))
```

Out[6]:
```
[1, 6]
```

In [7]:
```
range(10)
```

Out[7]:
```
range(0, 10)
```

In [8]:
```
r = range(10)
r
```

Out[8]:
```
range(0, 10)
```

In [9]:
```
for i in r:
    print(i)
```

```
0
1
2
3
4
5
6
7
8
9
```

In [10]:
```
range(1,10, 3, 4) # only 3 arguments were taken
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
Cell In[10], line 1
----> 1 range(1,10, 3, 4) # only 3 arguments were taken

TypeError: range expected at most 3 arguments, got 4
```

Dictionary

In [11]:
```
d = {}
type(d)
```

Out[11]:
```
dict
```

In [13]:
```
d1 = {'a':1, 'b':2, 3:True, 4.3:1+2j}
d1
```

Out[13]:
```
{'a': 1, 'b': 2, 3: True, 4.3: (1+2j)}
```

In [14]:
```
d1.keys()
```

Out[14]:
```
dict_keys(['a', 'b', 3, 4.3])
```

In [15]:
```
d1.values()
```

Out[15]:
```
dict_values([1, 2, True, (1+2j)])
```

In [16]:
```
d1
```

Out[16]:
```
{'a': 1, 'b': 2, 3: True, 4.3: (1+2j)}
```

In [17]:
```
d1.items()
```

Out[17]:
```
dict_items([('a', 1), ('b', 2), (3, True), (4.3, (1+2j))])
```

In [18]:
```
d2 = d1.copy()
```

In [19]:
```
d2
```

Out[19]:
```
{'a': 1, 'b': 2, 3: True, 4.3: (1+2j)}
```

In [20]:
```
d1 == d2
```

Out[20]:
```
True
```

In [21]:
```
print(d1)
print(d2)
```

```
{'a': 1, 'b': 2, 3: True, 4.3: (1+2j)}
{'a': 1, 'b': 2, 3: True, 4.3: (1+2j)}
```

In [22]:
```
d1
```

Out[22]:
```
{'a': 1, 'b': 2, 3: True, 4.3: (1+2j)}
```

In [23]:
```
d1[3]
```

Out[23]: True

In [24]:
```
d1[2]
```

```
---------------------------------------------------------------------------
KeyError                                  Traceback (most recent call last)
Cell In[24], line 1
----> 1 d1[2]

KeyError: 2
```

In [25]:
```
mydict3 = {1:'one', 2:'two', 'A':{'Name':'Mehar', 'Age':40}, 'B':('Bat', 'cat', 'hat')}
mydict3
```

Out[25]:
```
{1: 'one',
 2: 'two',
 'A': {'Name': 'Mehar', 'Age': 40},
 'B': ('Bat', 'cat', 'hat')}
```

In [26]:
```
mydict3
```

Out[26]:
```
{1: 'one',
 2: 'two',
 'A': {'Name': 'Mehar', 'Age': 40},
 'B': ('Bat', 'cat', 'hat')}
```

In [27]:
```
keys = {'a', 'b', 'c', 'd'}
mydict3 = dict.fromkeys(keys) # Create a dictionary from a sequence of keys
mydict3
```

Out[27]:
```
{'d': None, 'b': None, 'a': None, 'c': None}
```

In [28]:
```
keys = {'a', 'b', 'c', 'd'}
value = 10

mydict3 = dict.fromkeys(keys, value)
mydict3
```

Out[28]:
```
{'d': 10, 'b': 10, 'a': 10, 'c': 10}
```

In [30]:
```
keys = {'a', 'b', 'c', 'd'}
value = 10

mydict3 = dict.fromkeys(keys, value)
mydict3
```

Out[30]:
```
{'d': 10, 'b': 10, 'a': 10, 'c': 10}
```

In [31]:
```
keys = {'a' , 'b' , 'c' , 'd'}
value = [10, 20, 30]

mydict3 = dict.fromkeys(keys , value)
mydict3
```

Out[31]:
```
{'d': [10, 20, 30], 'b': [10, 20, 30], 'a': [10, 20, 30], 'c': [10, 20, 30]}
```

In [32]:
```
value.append(40)
mydict3
```

Out[32]:
```
{'d': [10, 20, 30, 40],
 'b': [10, 20, 30, 40],
 'a': [10, 20, 30, 40],
 'c': [10, 20, 30, 40]}
```

In [33]:
```
d1
```

Out[33]:
```
{'a': 1, 'b': 2, 3: True, 4.3: (1+2j)}
```

In [34]:
```
d1.get('a')
```

Out[34]:
```
1
```

In [35]:
```
mydict3
```

Out[35]:
```
{'d': [10, 20, 30, 40],
 'b': [10, 20, 30, 40],
 'a': [10, 20, 30, 40],
 'c': [10, 20, 30, 40]}
```

In [36]:
```
mydict3.get('d')
```

Out[36]:
```
[10, 20, 30, 40]
```

In [37]:
```
d1
```

Out[37]:
```
{'a': 1, 'b': 2, 3: True, 4.3: (1+2j)}
```

In [38]:
```
d1[10] = 'ten'
```

In [39]:
```
d1
```

Out[39]:
```
{'a': 1, 'b': 2, 3: True, 4.3: (1+2j), 10: 'ten'}
```

In [40]:
```
d1.pop(5)
```

```
---------------------------------------------------------------------------
KeyError                                  Traceback (most recent call last)
Cell In[40], line 1
----> 1 d1.pop(5)

KeyError: 5
```

In [41]:
```
d1.pop(4.3)
```

Out[41]:
```
(1+2j)
```

In [42]:
```
d1
```

Out[42]:
```
{'a': 1, 'b': 2, 3: True, 10: 'ten'}
```

In [43]:
```
d1.popitem()
```

Out[43]:
```
(10, 'ten')
```

In [44]:
```
d1
```

Out[44]:
```
{'a': 1, 'b': 2, 3: True}
```

In [45]:
```
all(d1)
```

Out[45]: True

In [46]:
```
any(d1)
```

Out[46]: True

In [47]:
```
d1[0] = 0
```

In [48]:
```
d1
```

Out[48]: {'a': 1, 'b': 2, 3: True, 0: 0}

In [49]:
```
all(d1)
```

Out[49]: False

In [50]:
```
any(d1)
```

Out[50]: True

PYTHON INBUILD DATASTRUCTURE COMPLETED